GAME DESIGN & LEVEL SPECIFICATION DOCUMENT (GDD)

An Echo Has No Shape

First-Person Acoustic Exploration & Regional Gating Design Specification

Author / Role	Kevin (Lefi) Shan: Solo Game & Level Designer, Programmer	**Engine & Stack**	Unity 6 (URP), C#, Custom HLSL Spherical Wave Shaders
Production Scope	4-Week Solo Prototype · USC CTIN (Situated Role: The Traveler)	**Core Constraints**	1 Primary Action (Press E) · Zero HUD / Minimaps · ~550m Single Scene

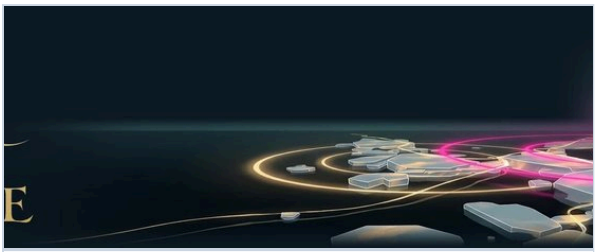


Figure 1.1: Player's white echo pulse colliding with the Siren's magenta wave in the Glacial Ocean.

Executive Summary & Level Design Goal

Many echolocation games treat sound as a simple flashlight in the dark: once the pulse reveals the room geometry, the mechanic stops evolving because the relationship between player input and returned information stays fixed.

To focus a 4-week solo cycle on level design and puzzle variety, An Echo Has No Shape keeps the player on a single primary action (Press E to Emit Echo, plus F to Interact) across a continuous ~550m x 550m fog basin, driving progression by changing how each region responds to that pulse.

1. Core Level Design Pillars & Acoustic Parameters

Section 1.1 to 1.3

1.1 Exploration Through Pacing (Traveler)

Rather than piecing together text logs, the player reads the world through positioning and timing. Where you stand when you emit a pulse, and when you pause to listen, directly control which paths open.

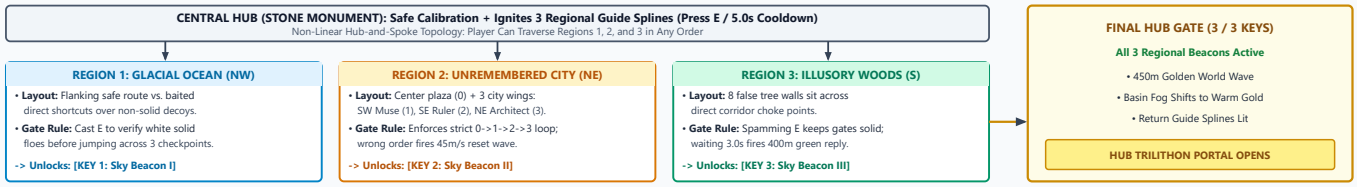
1.2 Pulse Range & 5.0s Cooldown Window

Pressing E emits a white wave that expands to 42m at 30 m/s and fades out at 1.4s on a 5.0s cooldown. The 3.6s dark gap between fade-out and cooldown prevents constant illumination and creates the window used by regional puzzles.

1.3 Testing Player Habits by Region

Each spoke tests a specific habit formed around pressing E: jumping onto any lit platform (Region 1), activating landmarks in arbitrary order (Region 2), and pressing E immediately whenever the cooldown finishes (Region 3).

Figure 1.2: Single-Verb Recontextualization & Regional Gating Progression (Hub -> Regions 1/2/3 -> Finale) Press E -> 3 Regional Rulesets -> 3/3 Beacons



2. Regional Level & Puzzle Rule Matrix

Table 2.1: Regional Rulesets & Gating

Design Axis	Hub: Stone Monument	Region 1: Glacial Ocean	Region 2: Unremembered City	Region 3: Illusory Woods
Level Topology	**Central Hub Ring** 3 spoke heads + final gate	**Branching Floe Lanes** Flanking arc vs. direct shortcut	**3-Wing Street Grid** Center plaza + SW / SE / NE wings	**Corridor Maze** Choke gates vs. detour loops
Habit Tested	Navigating without permanent sight or UI markers.	Jumping onto every lit platform along a shortcut.	Activating landmarks in random order or leaving early.	Spamming E immediately whenever the 5.0s cooldown ends.
Environmental Signal	**Prophecy Tablet** Ignites 3 guide splines	**3 Siren Emitters** 62m pink wave every 7.0s	**3 Wing Statues** 150m blue / red / green waves	**Forest Reply Center** 400m green wave after 3.0s wait
Fail State / Reset	Safe zone (zero hazards or reset states).	Stepping on a pink decoy floe respawns at last tablet.	Wrong order or exiting bounds fires 45m/s reset wave.	Pressing E early overwrites green wave & shuts gates.
Tablet Clue	*"Three silences wait in the mist..."*	*"Not every voice across the ice wishes you to reach the shore."*	*"The city stands whole within the meeting of many eyes."*	*"The woods have opened to the voice you left room to hear."*

3. Hub-and-Spoke World Layout & 4-Layer Sightless Wayfinding System

`~550m x 550m Single-Scene Basin (Main.unity)`

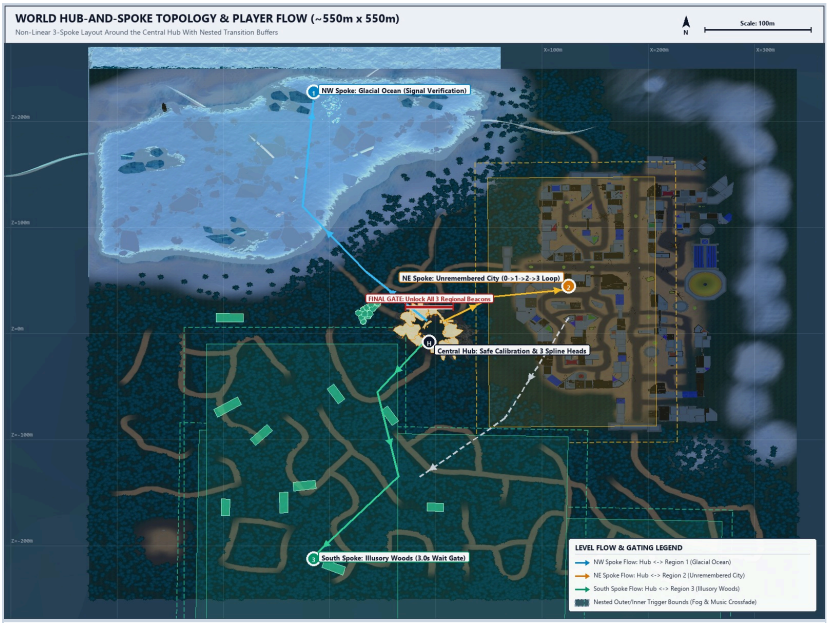


Figure 3.1: World Hub-and-Spoke Flow & Wayfinding Map (~550m x 550m): Showing the Central Hub (H), the 3 spoke traversal flows (1, 2, 3), secondary inter-regional loops, and nested fog transition bounds.

3.1 Non-Linear Hub-and-Spoke Routing

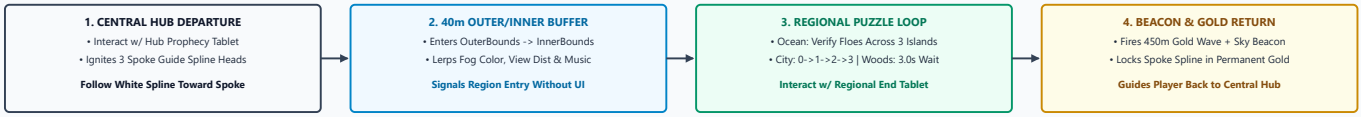
The ~550m x 550m basin is built inside a single continuous scene around the Central Hub. After interacting with the hub tablet to light the three guide spline heads, players can tackle the three spokes in any order:

- **NW Spoke (Glacial Ocean):** Leads from the western shore across three checkpoint islands to the northern beacon.
- **NE Spoke (Unremembered City):** Crosses the west bridge into a 200m x 220m walled street grid with an outer reset boundary.
- **South Spoke (Illusory Woods):** Has two entry points: the NW hub path (Tablet T1) and a secondary SE path from the city outskirts (Tablet T2).

3.2 Spoke Return Loop & Secondary Connectors

Completing any region ignites a vertical Sky Beacon above the fog and locks its guide spline into warm gold, creating a clear return trail to the Hub. Players can also traverse directly between regions via secondary connectors (such as the SE bridge linking the Unremembered City outskirts to the Illusory Woods).

Figure 3.2: Spoke Traversal & Nested Trigger Boundary Flow (Hub <-> 40m Buffer <-> Regional Core) 40m Buffer Crossfades Fog & Music



3.3 Four-Layer Wayfinding Specification Table

`Table 3.1: Spatial Orientation Layers`

Wayfinding Layer	Range	Level Design Mechanism	Technical Implementation	Player Navigation Purpose
1. Macro Scale	`150m to 450m`	**Sky Beacons & World Wave**	Completing a region fires a 450m gold wave (12s hold) and activates its sky beacon.	Gives players a high-altitude landmark visible above the fog from anywhere in the basin.
2. Meso Scale	`50m to 150m`	**Nested Fog & Audio Bounds**	Blends fog color, view distance, and music tracks across the 40m Outer/Inner buffer.	Tells the player they are entering or leaving a region before geometry appears.
3. Micro Scale	`15m to 50m`	**Breadcrumb Guide Splines**	Spline ribbons stay invisible until hit by an echo, then stream particles forward.	Guides players between major checkpoints without cluttering the ground with permanent arrows.
4. Immediate Scale	`0m to 15m`	**Anchor Silhouettes vs. Echo Terrain**	Splits materials into Seen (faint silhouette in fog) and Unseen (0% visibility until echoed).	Keeps key checkpoints (tablets, statues, spires) visible even while E is on cooldown.



Figure 3.3: Hub Initial State (0/3 Beacons): Player echo illuminating the monolith ring and central tablet.

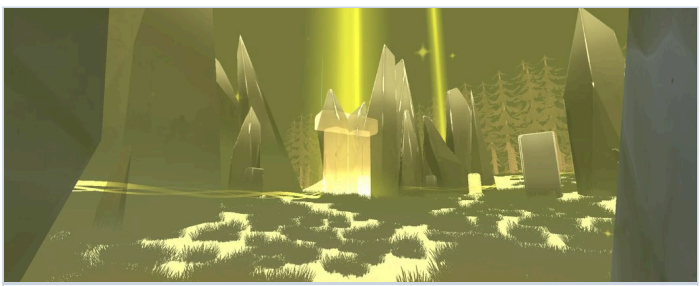


Figure 3.4: Hub Finale State (3/3 Beacons): All 3 regional beacons active and the Golden Gate unlocked.

4.1 Region 1 (Glacial Ocean): Flanking Safe Routes vs. Baited Decoy Shortcuts

`Signal Verification Before Jumping`

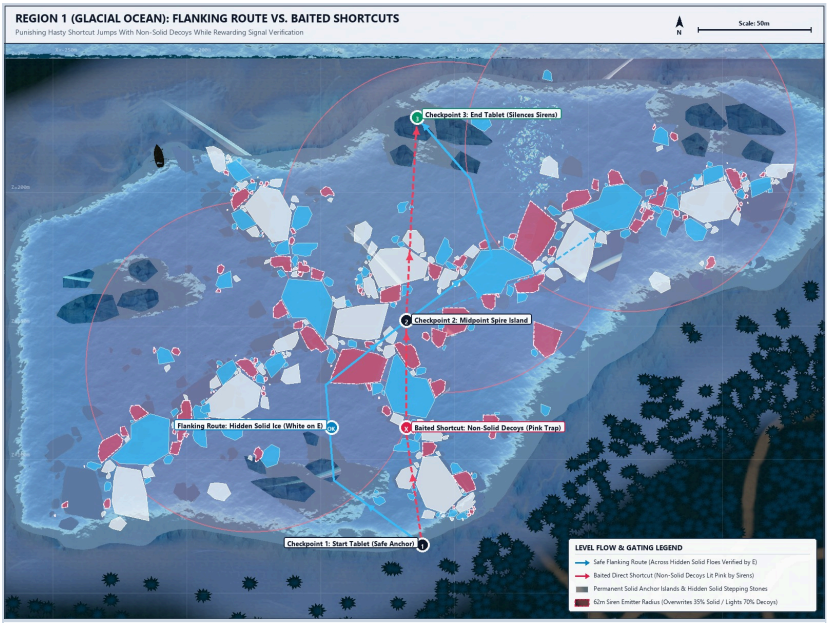


Figure 4.1: Region 1 Flow & Platform Classification Map (Glacial Ocean): Showing the Safe Flanking Route (blue arrows) across solid/hidden-solid ice vs. the Baited Direct Shortcuts (red dashed arrows) over non-solid Siren decoys.

Level Design Goal: Decision Tension Over Jump Precision

First-person platforming over water becomes repetitive if every visible platform is safe, and frustrating if jumps require tight mechanical precision. The Glacial Ocean builds tension through route verification: three Siren emitters (A, B, C) broadcast a 62m magenta wave every 7.0s that scrambles nearby platform colors and highlights non-solid decoy floes along direct shortcuts.

Three Platform Categories

- **1. Permanent Anchor Islands (White on Map):** Checkpoints 1 (Start), 2 (Twin Spires Midpoint), and 3 (End Beacon) retain a faint silhouette in the fog as safe observation islands.
- **2. Hidden Solid Floes (Cyan on Map):** Solid stepping stones arranged along the outer flanking arc. Invisible by default, they always turn pure white when swept by the player's E echo.
- **3. Non-Solid Decoy Traps (Pink Dashed on Map):** Collider-free water hazards placed along the shortest line between islands. Player echoes pass straight through them, while Siren waves light them pink.

Why the Shortcut Baits Impatient Players

Because the Siren wave tints some solid floes pink while lighting decoy floes in the exact same pink hue, players cannot trust magenta geometry. Pausing on an anchor island and pressing E reveals which platforms return a white outline, guiding the player around the safe flanking arc.

Figure 4.2: Region 1 Lane Routing & Platform Signal Verification Matrix

`Verify Pink Platforms With White Echo (Press E)`



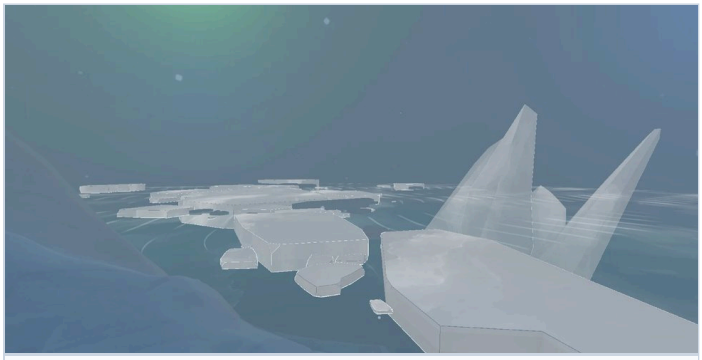


Figure 4.3: White Echo Verification: Player echo revealing solid white stepping stones before the twin spires.



Figure 4.4: Region 1 Completion State: Interacting with Checkpoint 3 silences the Sirens and lights the route in gold.

4.2 Region 2 (Unremembered City): 3-Wing Sequential Loop & Reset Boundary Gate `0 -> 1 -> 2 -> 3 Architectural Layering`

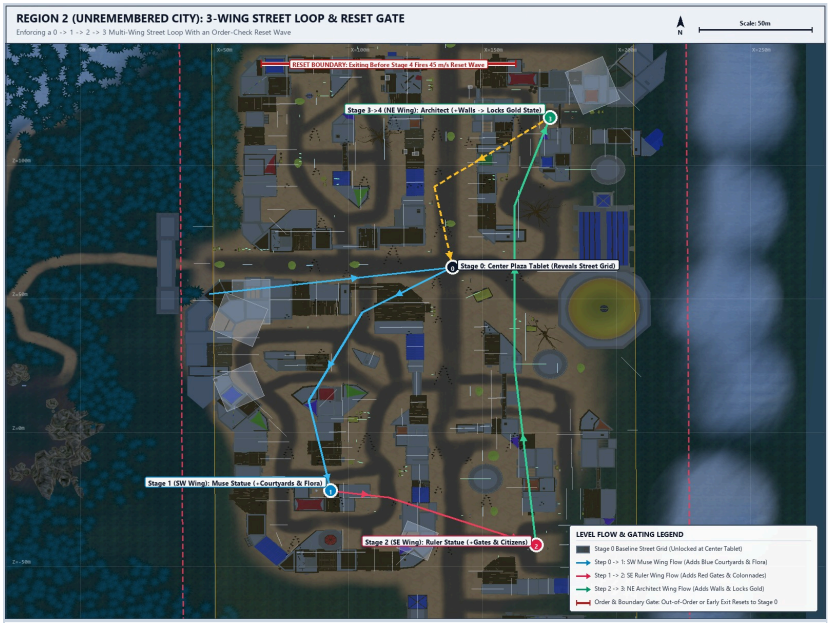


Figure 4.5: Region 2 Street Loop & Reset Boundary Map (Unremembered City): Showing the required street traversal loop from Center Plaza (0) -> SW Muse Wing (1) -> SE Ruler Wing (2) -> NE Architect Wing (3) and the outer Reset Boundary.

Level Design Goal: Multi-Wing Spatial Loop & State Layering

In the Unremembered City, each landmark reveals only one architectural layer of the ruined street grid. No single statue restores the city on its own. Instead of letting players activate the nearest statue and leave, the layout separates the three statues across the SW, SE, and NE wings around the central plaza.

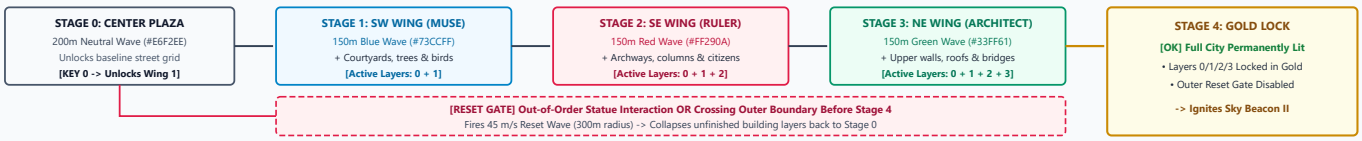
Additive Shader Layering Across City Wings

Normally, triggering a new environmental wave replaces the previous state. In Region 2, activating the Center Plaza Tablet (Stage 0) and then visiting the SW, SE, and NE statues in order (1 -> 2 -> 3) keeps every prior stage's geometry active. As the player walks from wing to wing, barren foundations gain courtyard trees, then colonnades and citizen silhouettes, and finally upper roofs and bridges.

Why the Outer Reset Boundary Exists

Because the SE city outskirts connect directly to the Illusory Woods, players could otherwise wander out of the city mid-puzzle. Surrounding the 200m x 220m grid with a reset boundary ensures that leaving before Stage 4 (or activating statues out of order) fires a 45 m/s Reset Wave from the center plaza, teaching the player that the city must be completed as one continuous loop.

Figure 4.6: Region 2 Lock-and-Key Sequence Chart (0 -> 1 -> 2 -> 3 -> Gold Lock) & Reset Gate Each Wing Statue [KEY] Preserves Earlier Layers



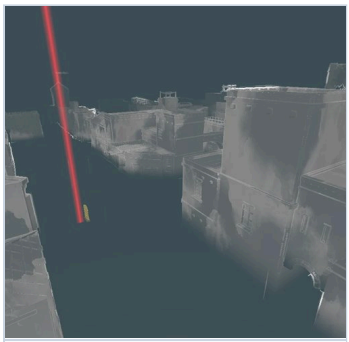


Figure 4.7: Stage 0 (Center): Baseline street grid and center tablet.

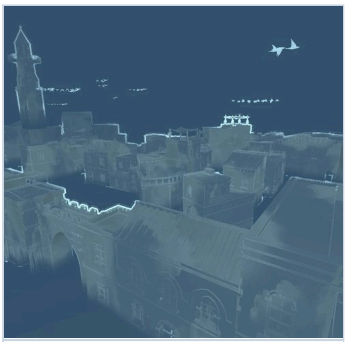


Figure 4.8: Stage 1 (SW Muse): Sky-blue fog, courtyards, and trees.



Figure 4.9: Stage 2 (SE Ruler): Crimson fog, archways, and citizens.



Figure 4.10: Stage 4 (Complete): All building layers locked in gold.

4.3 Region 3 (Illusory Woods): Corridor Choke-Point Gates & 3.0s Silence Window

`3.0s Delayed Forest Reply`

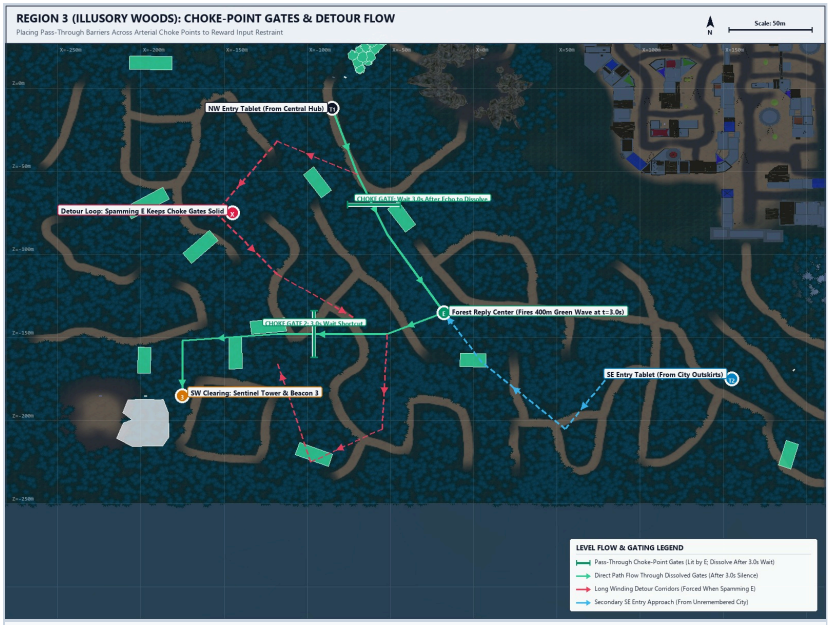


Figure 4.11: Region 3 Choke-Point Gates & Detour Flow Map (Illusory Woods): Showing the Direct Path (green arrows) through choke-point gates after a 3.0s wait vs. the Winding Detour Loops (red dashed arrows) when spamming E.

Choke-Point Gate Rule: Player Echoes Keep the Barrier Solid

Across the main junctions leading to the SW Sentinel Tower, 8 false tree walls sit across the dirt paths. When the player casts a white echo, the pulse lights up these false walls with the same white outline as solid corridor trees, making the direct path look like a dead end and funneling impatient players into long detour loops.

Unlocking the Choke Gates: If the player casts an echo and then waits 3.0s without pressing E again, the center of the forest answers with a 400m Green Reply Wave (50 m/s) that dissolves all false walls for 5.0s so the player can walk straight through. Pressing E during this open window triggers a 0.35s wave overwrite that cuts off the green wave and snaps the gates shut.



Figure 4.12: Corridor Spline: Dirt corridor lit by an echo-activated spline.

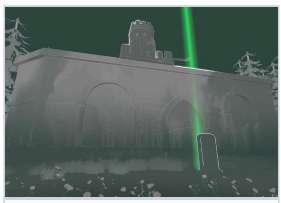


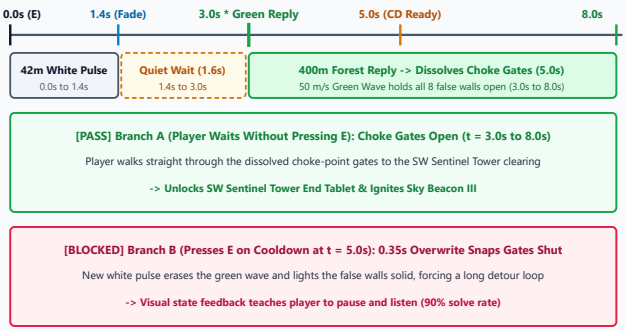
Figure 4.13: SW Sentinel Tower: Final clearing reached after passing the gates.

Table 4.1: Playtest Tuning of the 3.0s Wait Window (6 Cohorts)

Wait Time	Accidental	Solved Rate	Observed Playtest Behavior
`1.0s`	85%	15%	Players trigger green reply while walking without noticing why.
`2.0s`	45%	50%	Mixed results: fast walkers still trigger it by accident.
`3.0s` *	**10%**	**90%**	**[Tuned Sweet Spot] White pulse fades at 1.4s; leaves 1.6s quiet gap before 5.0s cooldown.**
`4.0s`	5%	60%	Impatient players start pressing E again just before cooldown ends.
`5.0s`	0%	20%	Collides with 5.0s cooldown: players press E on habit, resetting timer.
`6.5s`	0%	5%	Too long: players assume nothing happened and keep echoing.

Figure 4.14: 8s Input Timeline & Gate State

Wait 3.0s vs. Spam E



5. Level Design Retrospective & Key Takeaways

`Post-Mortem Synthesis`

5.1 Designing Around Cooldown Windows

Tuning the player's pulse to fade at 1.4s on a 5.0s cooldown created a 3.6s dark gap between casts, giving Siren waves (every 7.0s) and the Forest Reply (at 3.0s) a clear window to interact with level geometry.

5.2 Visual State Feedback Over Text Hints

In Region 3, tablet hints alone did not stop players from spamming E on cooldown. Adding the 0.35s wave overwrite (where pressing E visibly cuts off the green wave and snaps the gate shut in front of the player) raised solve rates from 15% to 90%.

5.3 Sightless Wayfinding Across a 550m Basin

Combining high-altitude sky beacons, 40m fog/audio crossfade buffers, echo-lit splines, and faint anchor silhouettes kept players oriented across a continuous ~550m scene without HUD markers or minimaps.